

केंद्रीय विद्यालय संगठन, अहमदाबाद संभाग
Kendriya Vidyalaya Sangathan, Ahmedabad Region
कक्षा दसवीं प्री-बोर्ड-I परीक्षा 2025-26

Pre Board I Examination 2025-26 for Class X
MATHEMATICS BASIC (241)

M.M.: 80

TIME: 3 Hours

GENERAL INSTRUCTIONS:

1. This question paper contains 38 questions. All Questions are compulsory.
2. This Question Paper is divided into 5 Sections A, B, C, D and E.
3. In Section A, Question numbers 1-18 are multiple choice questions (MCQs) and question no.19 and 20 are Assertion- Reason based questions of 1 mark each.
4. In Section B, Question numbers 21-25 are very short answer (VSA) type questions, carrying 02 marks each.
5. In Section C, Question numbers 26-31 are short answer (SA) type questions, carrying 03 marks each.
6. In Section D, Question numbers 32-35 are long answer (LA) type questions, carrying 05 marks each.
7. In Section E, Question numbers 36-38 are case study-based questions carrying 4 marks each with sub parts of the values of 1, 1 and 2 marks each respectively.
8. There is no overall choice. However, an internal choice in 2 questions of Section B, 2 questions of Section C and 2 questions of Section D has been provided. An internal choice has been provided in all the 2 marks questions of Section E.
9. Draw neat and clean figures wherever required. Take $\pi = 22/7$ wherever required if not stated.
10. Use of calculators is not allowed.

SECTION-A

- | | | |
|---|---|---|
| 1 | The exponent of 2 in prime factorization of natural number 288 is: | 1 |
| | (a) 5 (b) 4 (c) 3 (d) 2 | |
| 2 | If $2x + 3y = 5$ and $4x - 3y = 1$ then the value of $(x + y)$ is | 1 |
| | (a) 0 (b) -1 (c) 1 (d) 2 | |
| 3 | The zeroes of the quadratic polynomial $f(x) = x^2 + 99x + 127$ are | 1 |
| | (a) both negative (b) both positive | |
| | (c) both equal (d) can not be determined | |

- 4 The quadratic equation $2x^2 - \sqrt{3}x + 1 = 0$ has 1
 (a) two distinct real roots (b) two equal real roots
 (c) no real roots (d) more than 2 real roots

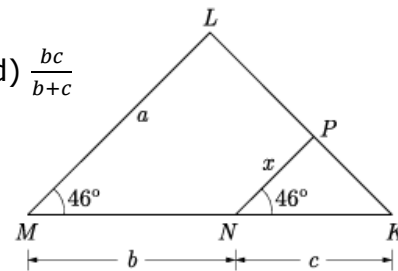
- 5 The sum of first 100 multiples of 3 is 1
 (a) 15130 (b) 15100 (c) 15120 (d) 15150

- 6 Which term of the AP 121, 117, 113,... is its first negative term? 1
 (a) $\frac{31}{4}$ (b) 29 (c) 31 (d) 32

- 7 $9\sec^2 A - 9\tan^2 A$ 1
 (a) 9 (b) 8 (c) 1 (d) 0

- 8 The coordinates of a point A, where AB is the diameter of a circle, whose centre is (-2,3) and B(4,8) is: 1
 (a) (-8,-2) (b) (0,-14) (c) (-8,10) (d) (-6,-2)

- 9 The value of x is 1
 (a) $\frac{ab}{a+b}$ (b) $\frac{ac}{a+c}$ (c) $\frac{ac}{b+c}$ (d) $\frac{bc}{b+c}$



- 10 ΔABC is such that $AB=3\text{cm}$, $BC = 2\text{cm}$, $CA = 2.5\text{ cm}$. If $\Delta ABC \sim \Delta DEF$ and $EF = 4\text{cm}$, then perimeter of ΔDEF is 1
 (a) 7.5cm (b) 15cm (c) 22.5cm (d) 30cm

- 11 The tangents drawn at the extremities of the diameter of a circle are 1
 (a) Perpendicular (b) Parallel
 (c) equal (d) unequal

- 12 If $\operatorname{cosec} \theta - \cot \theta = m$, then the value of $\operatorname{cosec} \theta + \cot \theta$ is 1
 (a) $1 - \frac{1}{m}$ (b) $\frac{1}{m}$ (c) $-m$ (d) $m^2 - 1$

- 13 The empirical relationship between the three measures of central tendency is 1
 (a) $3 \text{ Median} = \text{Mode} + 2 \text{ Mean}$ (b) $2 \text{ Median} = \text{Mode} + 2 \text{ Mean}$
 (c) $3 \text{ Median} = \text{Mode} + \text{Mean}$ (d) $3 \text{ Median} = \text{Mode} - 2 \text{ Mean}$

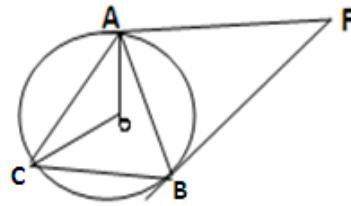
- 14 The following frequency distribution gives the monthly consumers of electricity a locality. 1

CI	85 -104	105 -124	125 -144	145 -164	165 - 184
Frequency	5	13	20	14	8

Find the difference between upper limit of modal class and lower limit of median class.

- (a) 0 (b) 1 (c) 8 (d) 9

- 22 In given figure O is center of the circum-circle of $\triangle ABC$. Tangents at A and B intersect P. If $\angle APB = 80^\circ$ and $\angle AOC = 140^\circ$ then find $\angle CAB$.



- 23 If α and β are zeroes of a quadratic polynomial $x^2 - 6x + 8$. Find the value of $\alpha^2 + \beta^2$. 2
- 24 Find the value(s) of k , if the quadratic equation $3x^2 - \sqrt{3} k x + 4 = 0$ has equal roots. 2

OR

Express the equation $\frac{1}{x} + \frac{1}{x+3} = \frac{9}{2}$ as a quadratic equation in standard form.

- 25 Evaluate: $\frac{\cos 45^\circ}{\sec 30^\circ + \operatorname{cosec} 30^\circ}$ 2

SECTION-C

- 26 Show that $7 - 2\sqrt{5}$ is irrational. 3

OR

Prove that $\sqrt{3}$ is irrational.

- 27 Find the median for the given frequency distribution: 3

Class	40-45	45-50	50-55	55-60	60-65	65-70	70-75
Frequency	2	3	8	6	6	3	2

- 28 Prove that length of the tangents drawn from an external point are equal. 3

- 29 How many natural numbers are there between 200 and 500 which are divisible by 8? 3

- 30 Prove that $(1 + \cot \theta - \operatorname{cosec} \theta)(1 + \tan \theta + \sec \theta) = 2$ 3

- 31 A part of monthly hostel charge is fixed and the remaining depends on the number of days one has taken food in the mess. When Swati takes food for 20 days, she has to pay Rs. 3,000 as hostel charges whereas Mansi who takes food for 25 days Rs. 3,500 as hostel charges. Find the fixed charges and the cost of food per day. 3

OR

The grocery store we use does not mark prices on its goods. My sister went to this store, purchased three 1-kg packages of almond and two 500-gram packages cashew, and paid a total of Rs 1345. Not knowing that she went to the store, I also went to the same store, purchased two 1-kg packages of almond and three 500-gram packages cashew, and paid a total of Rs 1145. Now we want to return two 1-kg packages of almond and two 500-gram packages cashew. How much will be refunded.

SECTION-D

- 32 (a) If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points then the other two sides are divided in the same ratio. State and Prove its converse. 5

OR

(b) It is given that sides AB and AC and median AD of ΔABC are respectively proportional to sides PQ and PR and median PM of another ΔPQR . Show that $\Delta ABC \sim \Delta PQR$.

- 33 A train travelling at a uniform speed for 360 km have taken 48 minutes less to travel the same distance if its speed were 5 km/hour more. Find the original speed of the train. 5
- 34 From the top of a building 75 meters high, the angles of depression to the top and bottom of a tower are observed to be 45° and 60° respectively. Find the height of the tower. Also, find the horizontal distance between the building and the tower. ($\sqrt{3} = 1.732$ m) 5
- 35 A spherical glass vessel has a cylindrical neck 7 cm long and 8 cm in diameter. The radius of spherical part is 10 cm. Find the volume of the vessel. 5

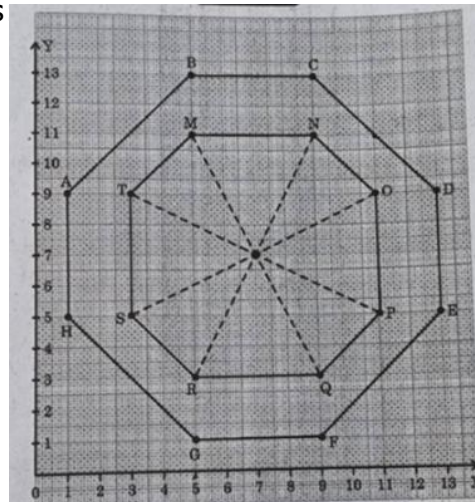
OR

A circus tent is in the shape of a cylinder surmounted by a conical top of same diameter. Their common diameter is 56 m and the height of cylindrical part is 6 m and the total height of the tent above the ground is 27 m, find the area of canvas used in the tent.

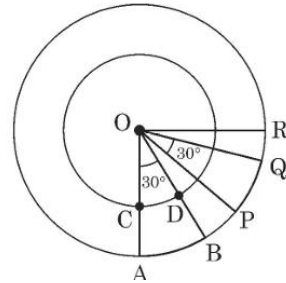
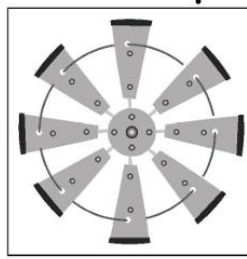
SECTION-E

- 36 The top of a table is hexagonal in shape. On the basis of the information given above, answer the following questions 4

- i) Write the coordinates of A and D.
- ii) Write the coordinates of the mid-point of line segment joining E & F.
- iii) Find the co-ordinates of the point which divides the line segment joining M and N in the ratio 1:3.



- 37 A farmer has put up a decorative windmill in his farm in which there are eight blades of equal width and equally placed in a circular arrangement. A circular wire goes through them. The diagram shows two blades OAB and OPQ in a quarter circle with centre O. $\angle AOB = \angle POQ = 30^\circ$, $OA = 28$ cm, $OC = 21$ cm. O is the centre of both the circles. 4



- (i) Determine the measure of $\angle BOP$
(ii) Find length of arc CD.
(iii) (a) Find the area of region CABD.
OR
(iii) (b) Find perimeter of region CABD.

- 38 Cards on which numbers 1, 2, 3.....100 are written (one number on one card and no number is repeated), put in a bag and are mixed thoroughly. A card is drawn at random from the bag. Find the following probability.
(i) What is the probability that card taken out has an odd number?
(ii) What is the probability that card taken out has a two-digit odd number?
(iii) What is the probability that card taken out has an odd number which is not less than 70?

OR

Computers in Elementary Schools: Computer-based learning (CBL) refers to any teaching methodology that makes use of computers for information transmission or input. At an elementary school level, computer applications can be used to display multimedia lesson plans or students can input examination answers into a computer to facilitate grading and evaluation. Multimedia formats and applications that allow students to pause content are especially valuable in early childhood education especially since students are coming to school with different competencies, learning styles and speeds for absorbing information.

Total 1000 elementary and secondary schools of Rajasthan were classified by the number of computers they had.

Computers	1-10	11 -20	21-50	51-100	>100
Schools	250	210	290	170	80

Choose one school at random and find the following probability.

- (i) Find the probability that it has 50 or fewer computers.
(ii) Find the probability that it has more than 100 computers.
(iii) Find the probability that it has no more than 20 computers.



1
1
2